Reading at Scale: Critical Reading, Metadata, and Argument in Digital History

These readings changed my thinking not by introducing more tools, but by making me question my earlier assumption that historical sources are already there and simply need to be found more quickly. I previously treated digitised newspapers as copies that widened access. I now understand them as new editions produced through collection decisions, microfilming, scanning, optical character recognition (OCR), metadata cataloguing, and interface ranking. A search box queries OCR text rather than the newspaper itself; a missing result does not prove historical absence. In my research on standards governing the production of Chinese medicines in China from 1910 to 1950, searches in *Shenbao* for terms such as “preparation methods” (*zhifa*), “medicinal processing” (*paozhi*), and “instruction leaflets” (*fangdan*) may be affected by older character forms, column layouts, and advertisements. The digital medium therefore shapes the evidence visible to me. Page numbers, sections, font sizes, and surrounding advertisements may also indicate the intended audience and claimed authority of a reform proposal (Salmi, 2020, pp. 46 – 49, 56, 67 – 68).

I therefore cannot treat results returned by a database as a ready-made dataset. Data curation is itself an initial argument about what can represent the historical problem. I will use *Shenbao* as a point of entry rather than as a proxy for the national picture, and compare it with newspapers from other regions, regulations, archival records, and materials produced by pharmaceutical manufacturers. I need to record corpus boundaries, issue and page numbers, search terms and their orthographic variants, OCR corrections, and reasons for inclusion or exclusion, while retaining page images. I would stratify spot checks by period and page layout, sampling both retrieved results and pages not retrieved by the search. This would allow me to estimate false positives and false negatives rather than checking only examples that fit my expectations. The selection, classification, and description of sources already contain implicit historical judgements (Robertson & Mullen, 2021, p. 1008).

This recognition has also changed how I evaluate digital methods. If I want to see when terms such as “improvement,” “science,” and “standard” clustered, word-frequency and co-occurrence analysis may identify relevant periods, but cannot determine what those words meant. The same applies to topic modelling: researchers choose the number of topics, and the output consists of word clusters requiring interpretation, not themes naturally present in the sources. To examine regional differences with GIS, I must first distinguish the newspaper’s place of publication, the location of an event, and the jurisdiction of a regulation. If the question concerns relationships between people and institutions, a network may be more suitable than a map, but a missing link shows only that the relationship was not recorded in the sources. Distant reading can identify concentrations, anomalies, and gaps; close reading must test their meaning and revise the initial hypothesis. Explaining why a tool is or is not appropriate demonstrates more

methodological judgement than producing more visualisations.

In future research, I will evaluate digital history outputs for traceability, transparency, argumentative strength, and accessibility, rather than judging them only by their visual presentation. A chart should link back to the original pages and disclose corpus boundaries, cleaning rules, parameters, errors, and versions. For example, Shanghai's apparent density on a map may only reflect the greater searchability of *Shenbao*. Lines may also misrepresent simultaneous appearances as evidence of transmission. Digital patterns must therefore be weighed against competing explanations and turned into historical arguments through close reading, other sources, and existing scholarship (Robertson & Mullen, 2021, pp. 1010 – 1018). However, I do not regard journal articles as the only valid scholarly outputs. Digital collections and public-facing projects also have scholarly value, provided that they make clear to users the implicit arguments created by selection and classification.

Paywalls and manual auditing show that ethical problems cannot be solved simply by asking individual researchers to be more responsible. Commercial subscription models, digitisation and licensing costs, inadequate funding for archives, and restrictions imposed by copyright and platform contracts produce unequal access. Manual audits are also constrained by data scale, short-term funding, unavailable raw OCR, and academic rewards favouring novelty over correction and maintenance (Salmi, 2020, pp. 44 – 50, 62 – 63; Robertson & Mullen, 2021, p. 1007). Institutions should budget for long-term maintenance, stratified audits, and multilingual OCR. Where rights permit, they should also publish metadata, collection boundaries, error rates, and versioned corrections. For my part, I should state the scope of access, search dates, excluded paywalled materials, and error estimates from my spot checks. I should not turn "inaccessible to me" into "historically nonexistent."

For me, the value of digital history lies not in processing sources faster, but in making the production of evidence more visible. Historians should not wait until the end of the process to interpret machine-generated results. At every stage—from searching, source selection, cleaning, and classification to visualisation and publication—historians make judgements that carry interpretive and ethical responsibilities.

References

Robertson, S., & Mullen, L. (2021). Arguing with digital history: Patterns of historical interpretation. *Journal of Social History, 54*(4), 1005 – 1022. https://doi.org/10.1093/jsh/shab015

Salmi, H. (2020). *What is digital history?* Polity Press.